STAT 992: Foundation Models for Biomedical Data

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Lecture 11: Prompts, In-Context Learning, Test-Time Adaptation

March 2, 2026

Logistics – Review Paper

- Review Paper Structure Updated

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7. Open Problems and Future Directions
8. Conclusions

Logistics - Lectures

- We have a few open slots of undecided lecture topics:
- Vote for lecture topics
  - Will use the top 3
- First seminar this Wednesday
  - Circuit Tracing (Jingyun) & PluRel (Jiaqi)
- Everyone: add questions/comments to notes docs

3/30	Lecture #16 : Spring Break [slides | notes]
4/6	Lecture #17 (Prof. Lengerich): **TBD** [slides | notes]
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4/13	Lecture #18 (Prof. Lengerich): **TBD** [slides | notes]
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4/27	Lecture #20 (Prof. Lengerich): **Open Problems & Paper Wrap-Up** [slides | notes]

Today: Test-Time Adaptation in Foundation Models

- Classic ML Pipeline:
  - Dataset → Training → Model → Prediction
- Foundation Model Pipeline:
  - Pretrained Model → Prompt → Prediction

Modern foundation models adapt primarily **at inference time rather than training time.**

Today: Test-Time Adaptation in Foundation Models

- We adapt model behavior without retraining by modifying the inference process.

Method	How
Prompting	modify context
In-Context Learning	provide demonstrations
Reasoning	generate intermediate tokens
RAG	inject external knowledge
Tools	interact with environment
Memory	persistent user context

These methods increase the **effective computation performed at inference time**.

Prompting

Prompt Design Matters

- **Bad Prompt**

Classify the sentiment.

Text: I loved the movie.

- **Good Prompt**

Classify the sentiment.

Text: I loved the movie.
Sentiment: Positive

Text: The acting was terrible.
Sentiment: Negative

Text: The plot was incredible.
Sentiment:

Types of Prompting

- zero-shot
- few-shot
- chain-of-thought
- tool-augmented
- structured prompts

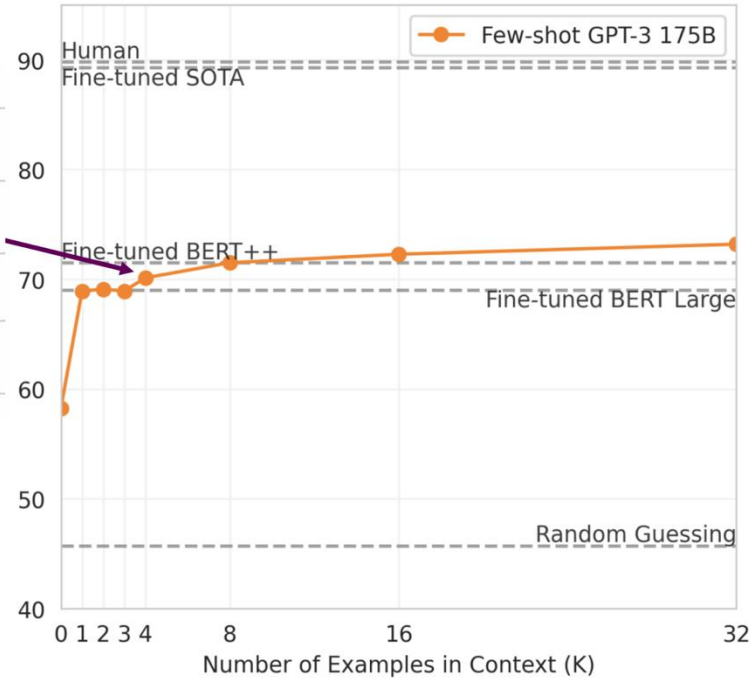
Prompting creates the **training distribution the model sees during inference.**

Emergence of Few-Shot Learning

- GPT-2 showed early **zero-shot** capability
- GPT-3 revealed strong **few-shot** performance
- Performance scales with model size

Few-shot

```
Translate English to French:
sea otter => loutre de mer
peppermint => menthe poivrée
plush girafe => girafe peluche
cheese =>
```



[Brown et al., 2020]

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Formatting matters

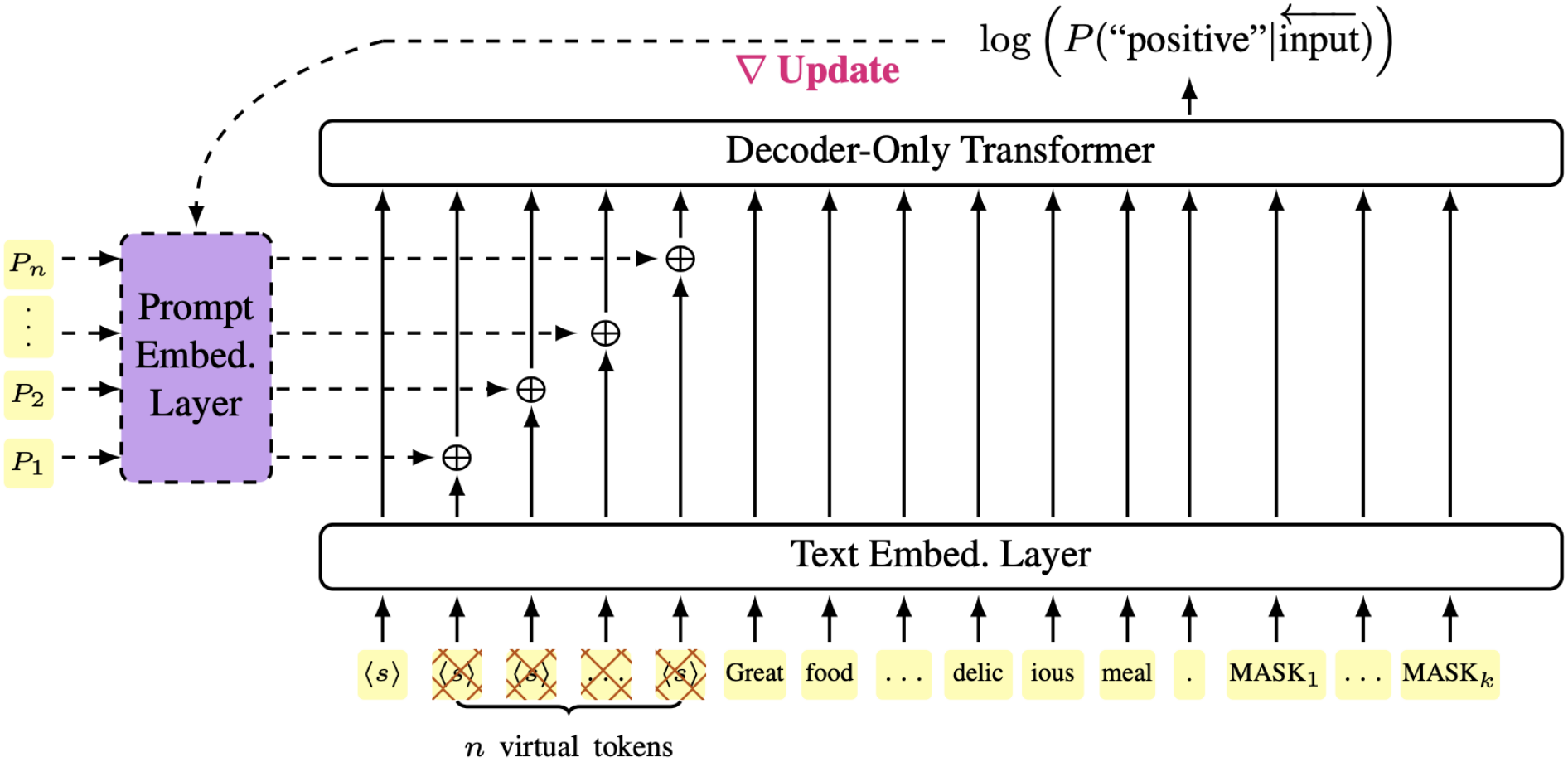
Models are sensitive to:

- example order
- label balance
- lexical similarity

This is because attention weights depend on token similarity.

Prompt Optimization

Soft Prompting: Optimization



https://arxiv.org/pdf/2306.04735

DSPy: Programming and Optimizing LLM Pipelines

- Key idea: Treat prompting as an **optimization problem over prompt programs**.
- Instead of manually writing prompts: prompt → model → output
- DSPy defines a **program with components, and optimizes the program**

DSPy *Programming—not prompting—LMs*

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downloads 27M downloads/month 4M

DSPy is a declarative framework for building modular AI software. It allows you to **iterate fast on structured code**, rather than brittle strings, and offers algorithms that **compile AI programs into effective prompts and weights** for your language models, whether you're building simple classifiers, sophisticated RAG pipelines, or Agent loops.

Instead of wrangling prompts or training jobs, DSPy (Declarative Self-improving Python) enables you to **build AI software from natural-language modules** and to *generically compose them* with different models, inference strategies, or learning algorithms. This makes AI software **more reliable, maintainable, and portable** across models and strategies.

tl;dr Think of DSPy as a higher-level language for AI programming, like the shift from assembly to C or pointer arithmetic to SQL. Meet the community, seek help, or start contributing via GitHub and Discord.

https://dspy.ai/

In-Context Learning

“In-Context Learning”

Few-shot

```
Translate English to French:
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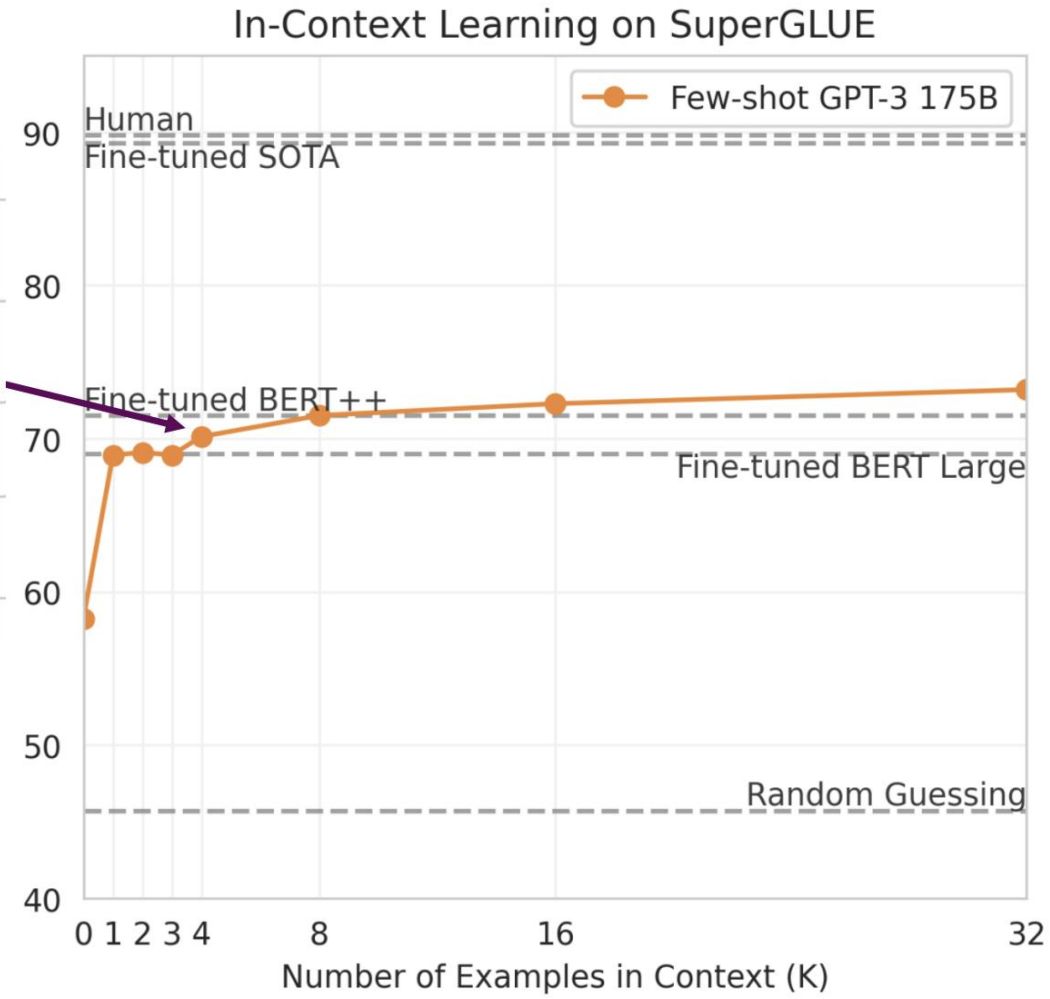

[Brown et al., 2020]

What is In-Context Learning Doing?

Possible explanations:

- nearest neighbor retrieval
- kernel regression
- pattern completion
- gradient-based learning

In-Context Learning as Gradient-Based Learning?

Suppose we have prefix containing training examples $(x_1, y_1), \ldots, (x_n, y_n)$, Query x_{n+1}, and goal y_{n+1}

<u>One Gradient-Descent Update</u>

$$y = w^T x$$

$$w_1 = w_0 - \eta \sum_i (w_0^T x_i - y_i) x_i$$

New prediction:

$$\hat{y}_{n+1} = w_0^T x_{n+1} + \eta \sum_i (x_{n+1}^T x_i)(y_i - w_0^T x_i)$$

<u>One Layer of Transformer Attention</u>

New prediction:

$$h = \sum_i \alpha_i v_i, \qquad \alpha_i = \mathrm{softmax}(q^T k_i)$$

$$v_i = f(x_i, y_i), \qquad q^T k_i = x_{n+1}^T W_q^T W_k x_i$$

$$h = \sum_i softmax(x_{n+1}^T W_q^T W_k x_i) f(x_i, y_i)$$

Gradient descent updates depend on similarity between the query and training examples. **Attention computes similarity-weighted mixtures of examples.** Because transformers can represent such computations, meta-training can induce gradient-descent-like update rules during the forward pass.

Scaling Test-Time Inference

Scaling Test-Time Inference

Accuracy can scale with **inference compute**, not just model size.

Test-time scaling techniques:

- Chain-of-thought reasoning
- Self-consistency sampling
- Tool use / search
- Sleep-time compute

Chain-of-Thought

Asking the model to generate reasoning tokens improves problem-solving accuracy.

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ✗

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✓

Wei, et al. (2023) Chain-of-Though Prompting Elicits Reasoning in LLMs

Sleep-time compute

Main idea: there is time in-between prompt inputs and the next prompt is fairly predictable; start inference before the next prompt.

https://arxiv.org/pdf/2504.13171

Letta

Sleep-time Compute: Beyond Inference Scaling at Test-time

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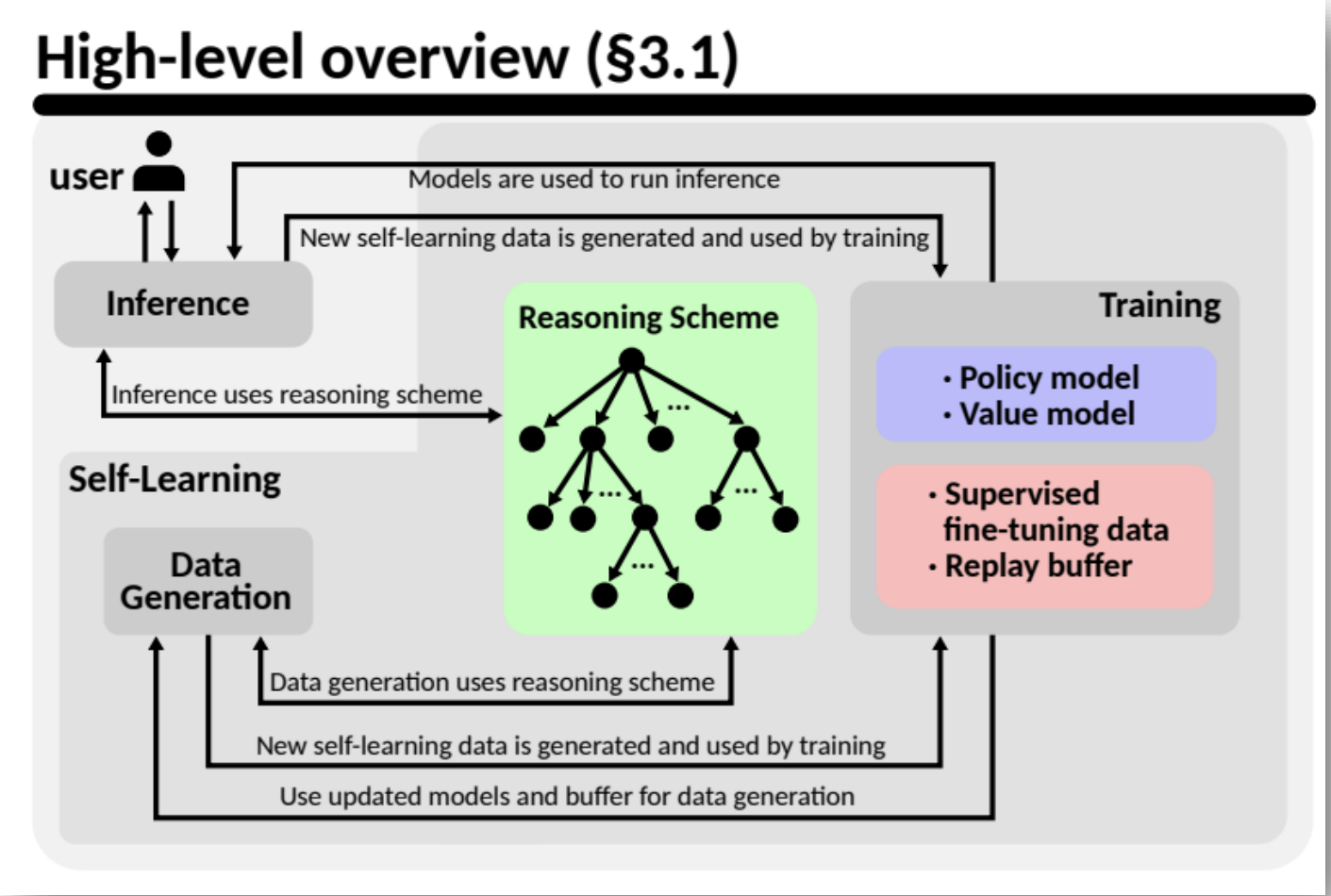
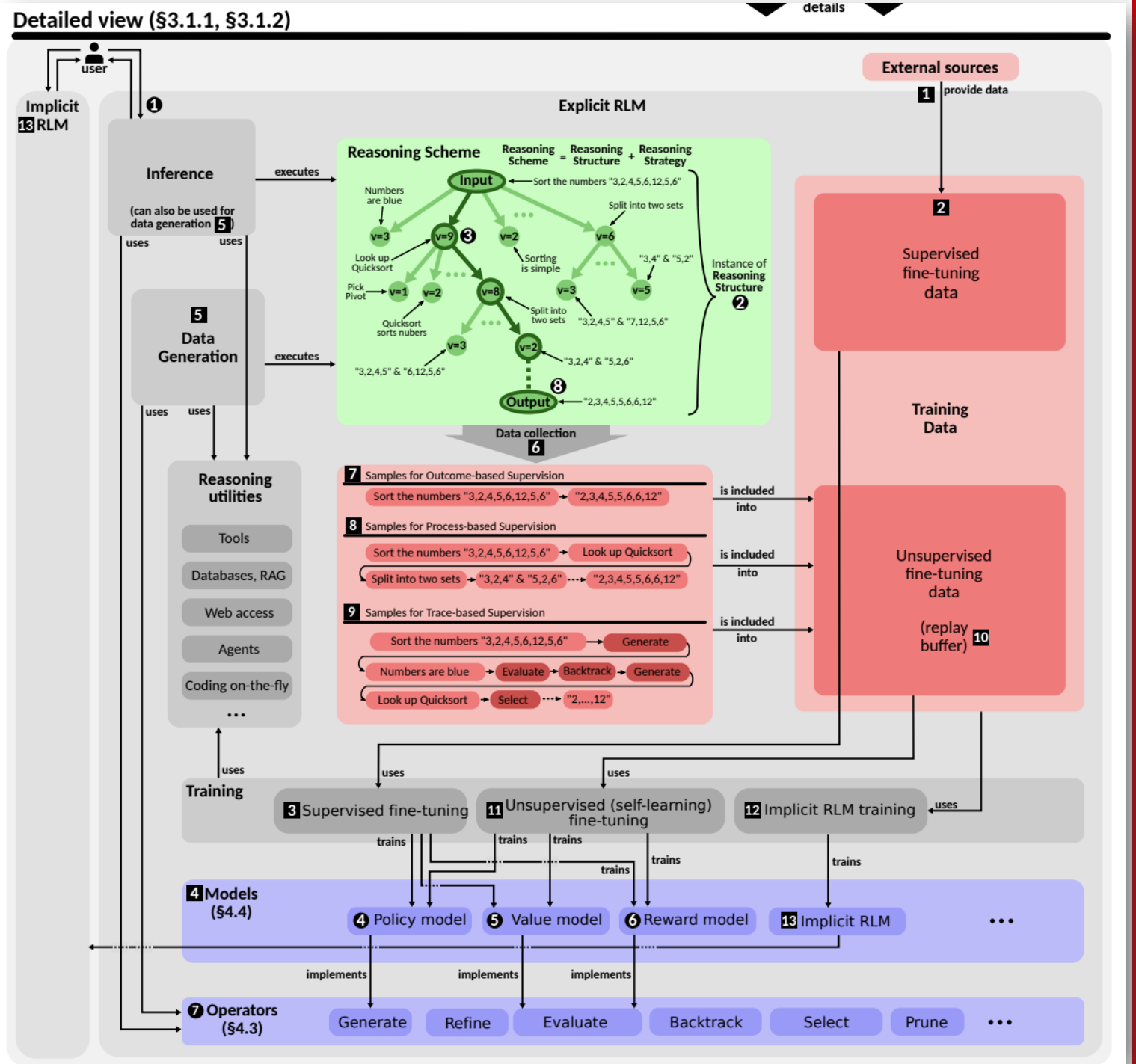
Reasoning Models

Reasoning models increase **test-time compute** by generating intermediate reasoning steps.



https://arxiv.org/pdf/2501.11223

Reasoning Models



https://arxiv.org/pdf/2501.11223

Parameter Efficient Fine-Tuning

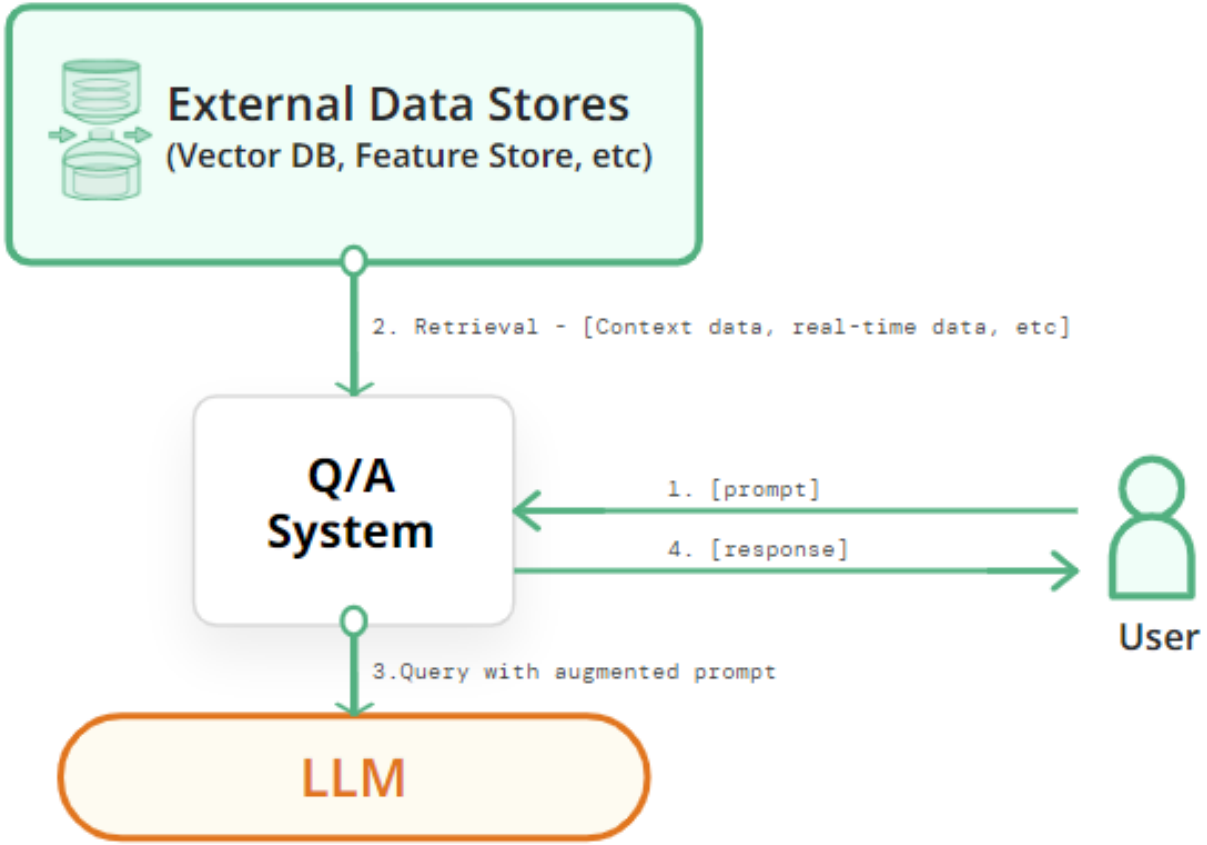
Personalization / Adaptation / Alignment

- Every user has their own preferences, history, and contexts.
- **How can we efficiently adapt to each user?**

- Methods:
  - Memory
  - RAG
  - User profiles
  - Preference learning

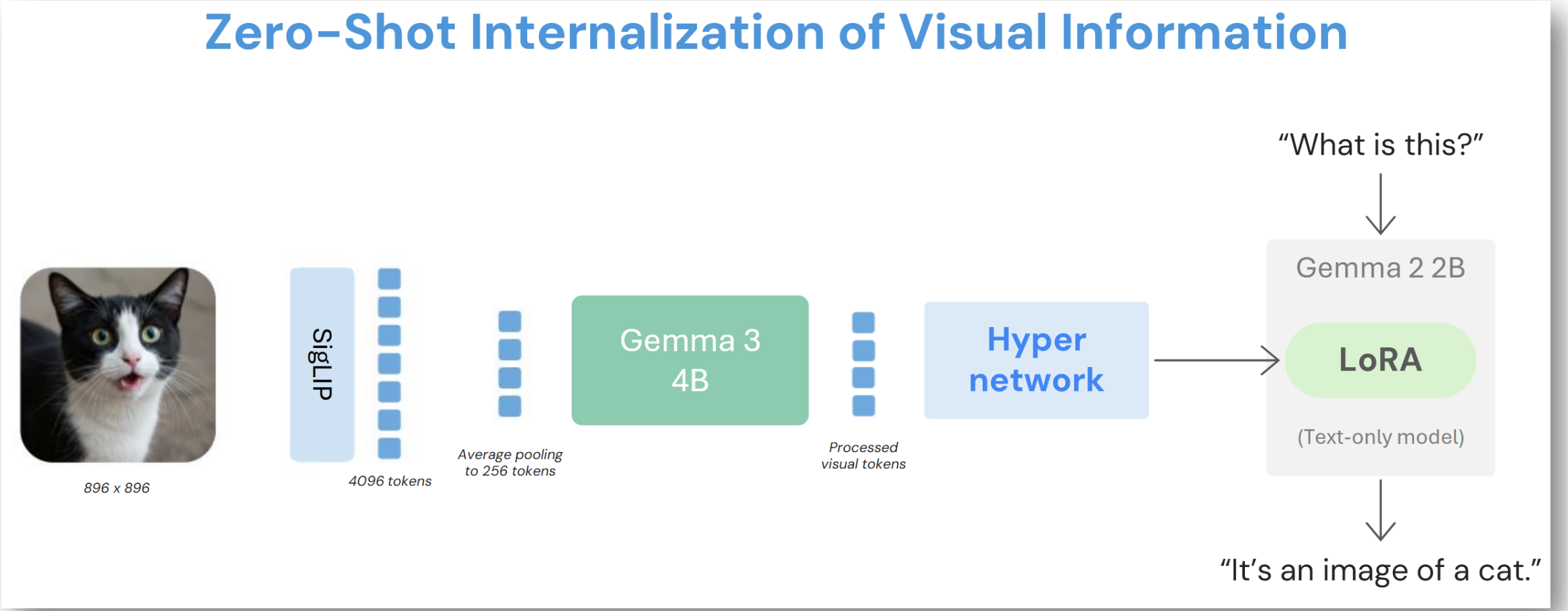
Retrieval-Augment Generation

- Resource access enables personalization
- **Key idea: knowledge retrieval + reasoning**
- Current skepticism:
  - LLMs **trust whatever you give them** (bad retrieval → hallucinated reasoning)
  - Embeddings are a weak proxy for relevance
  - Large context windows reduce the advantage of RAG



https://www.hopsworks.ai/dictionary/retrieval-augmented-generation-llm

RAG-to-LoRA?



https://pub.sakana.ai/doc-to-lora/

Questions?

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